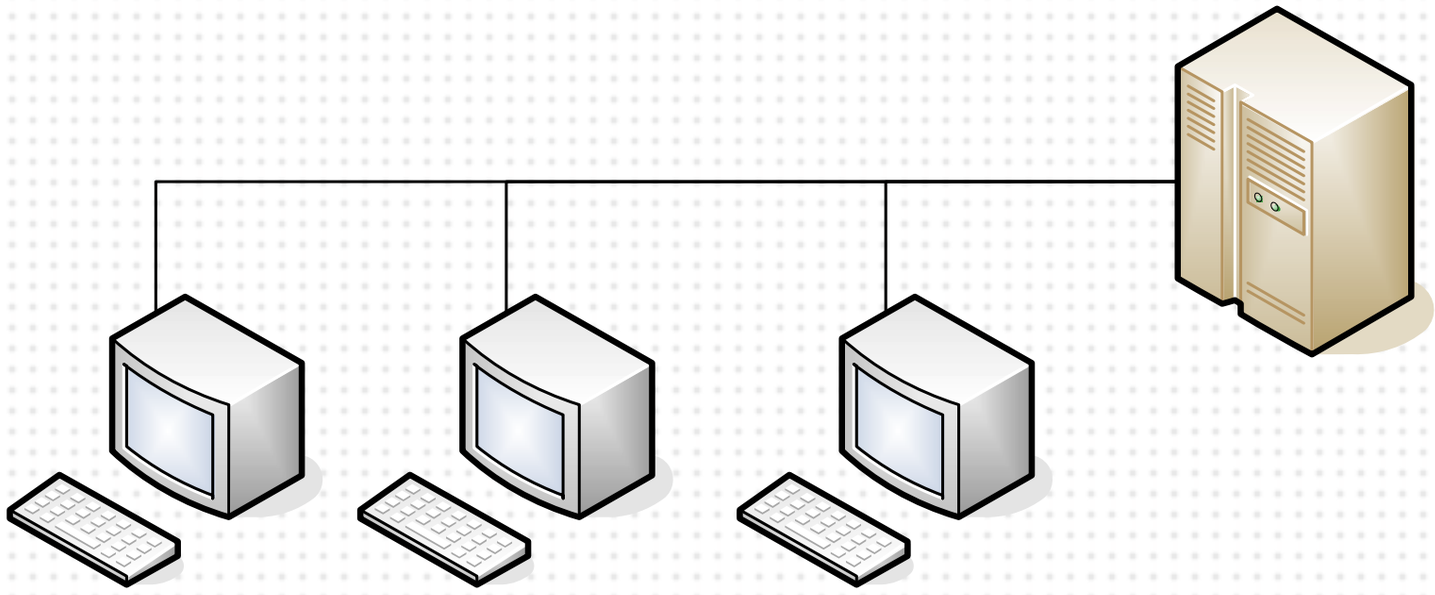
Single Page Applications (Rich Internet Application - RIA)

Ehsan Khamespanah

Spring 2026

The Early Days



terminals

only handle input and output

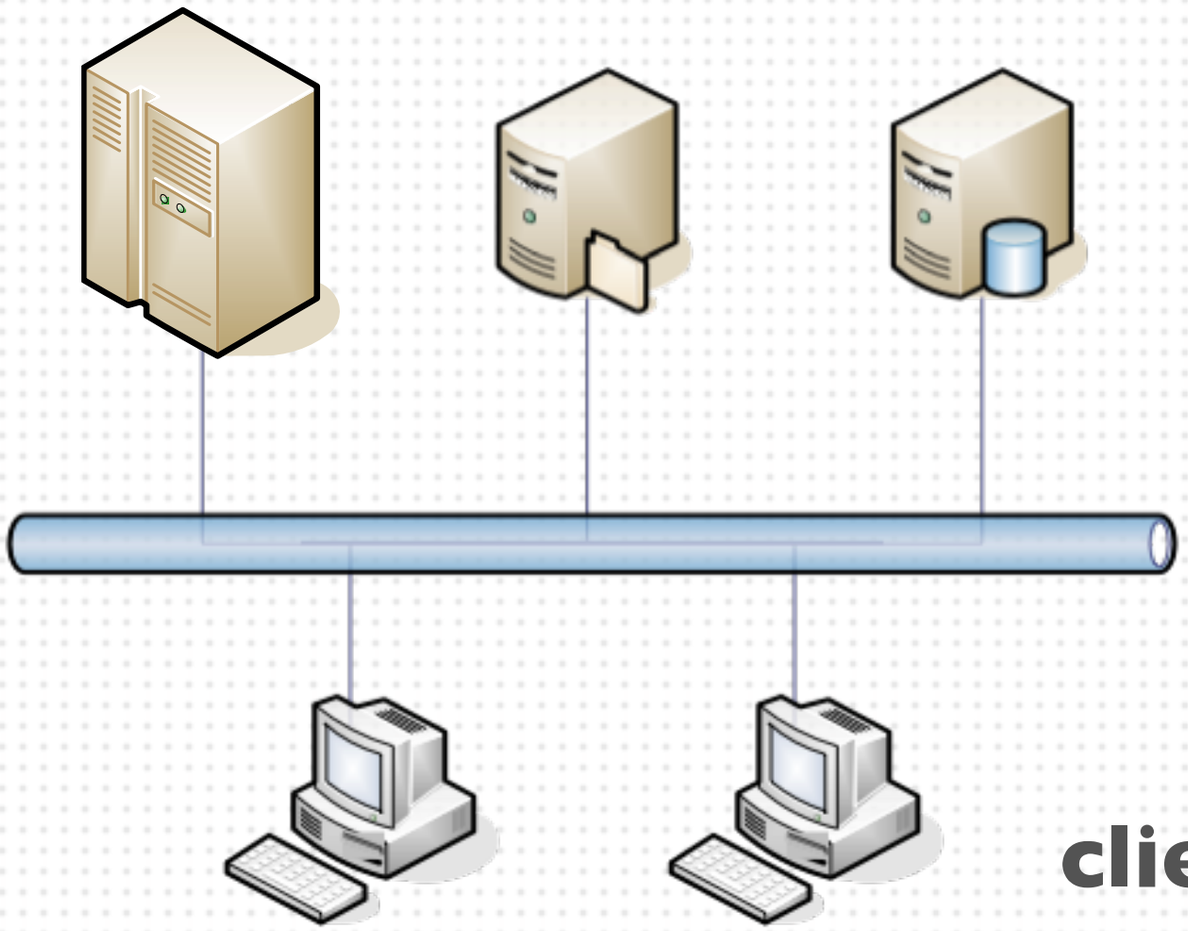
Very limited processing (e.g. line editing)

mainframe

handles all the processing

Capable of handling and processing very large amounts of data quickly

Client-Server Architecture



servers

provide various types of services (data, file, etc.)

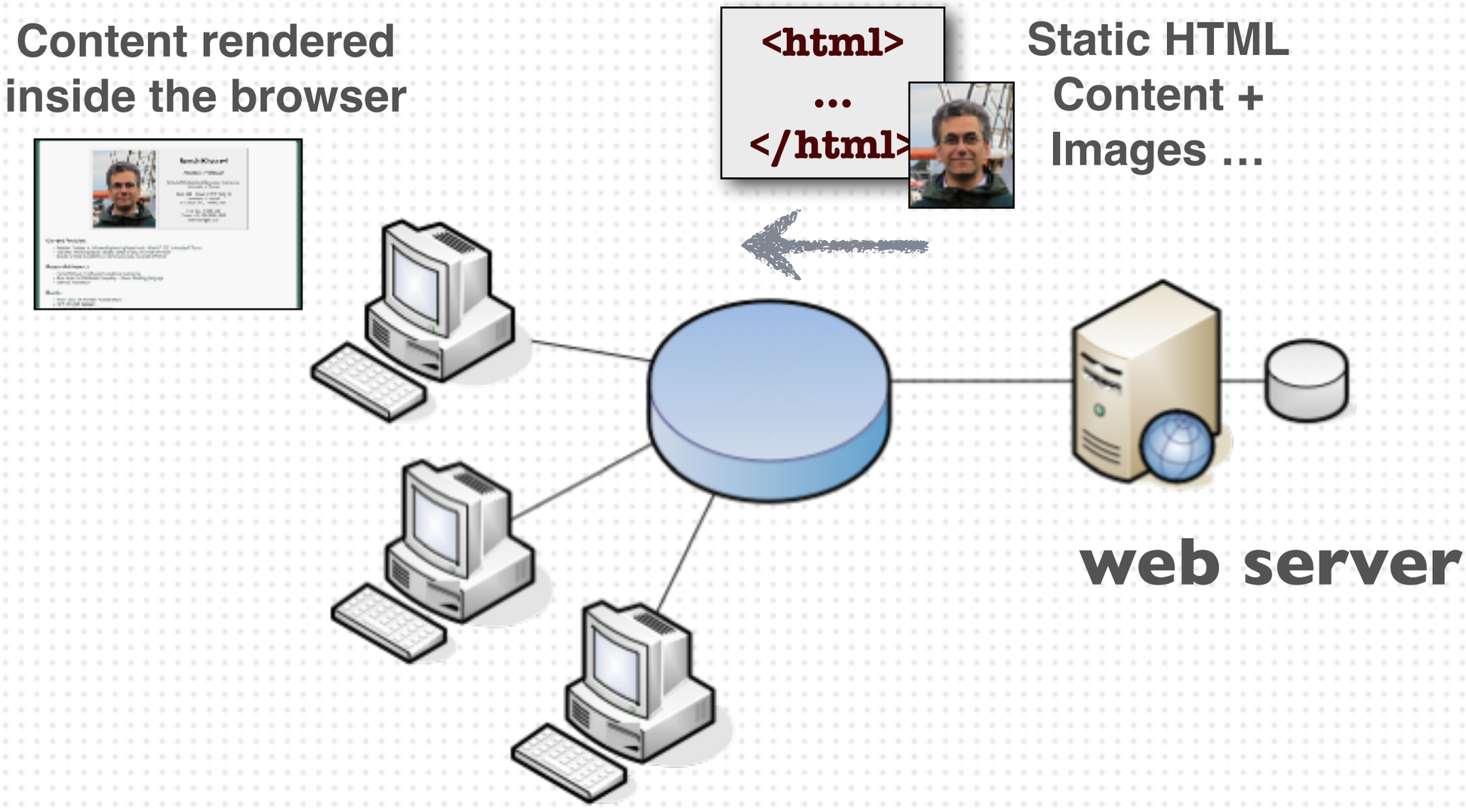
mainframe and mid-range computers

clients

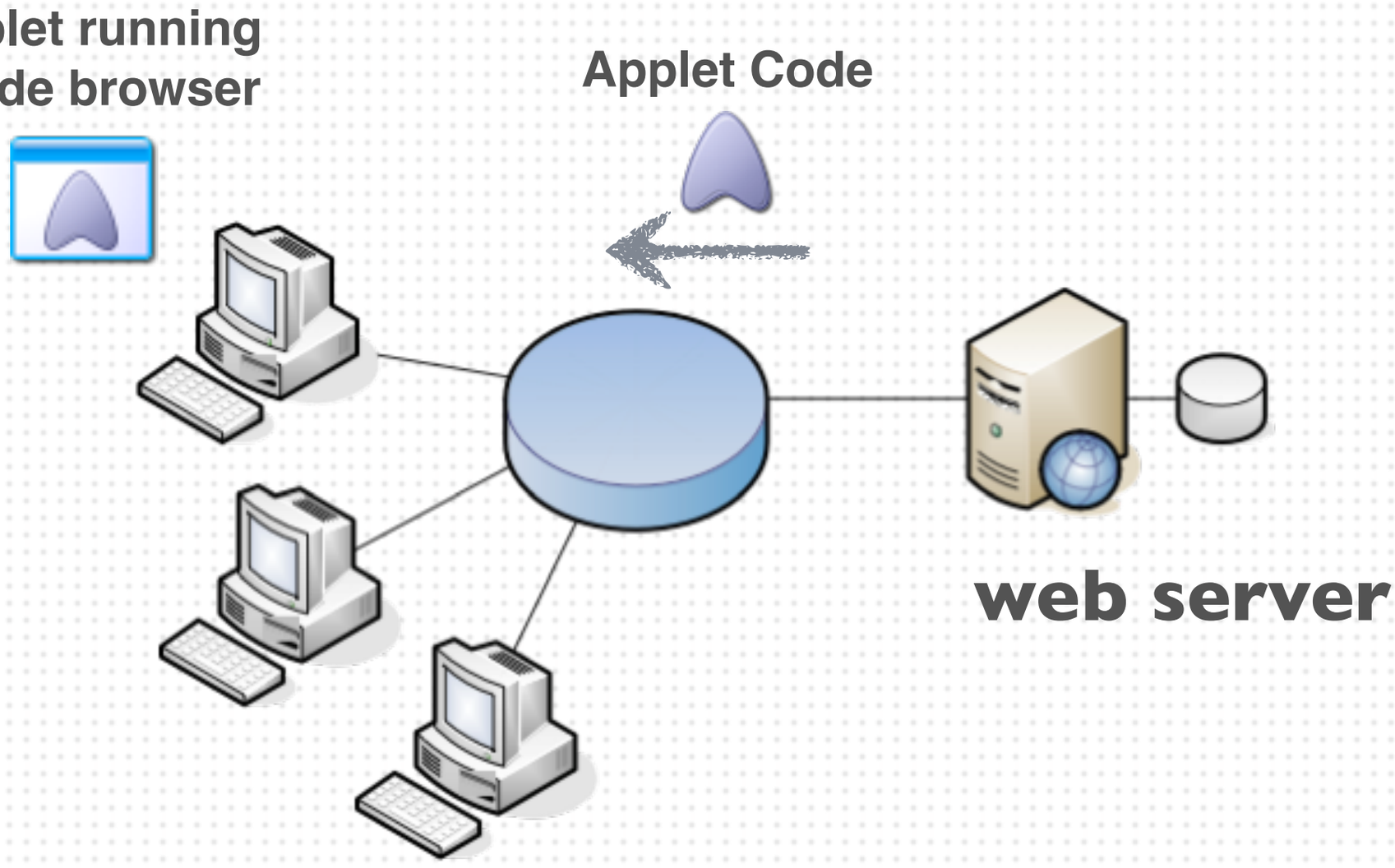
handle part of the processing

usually desktop PCs -- way more powerful than terminals

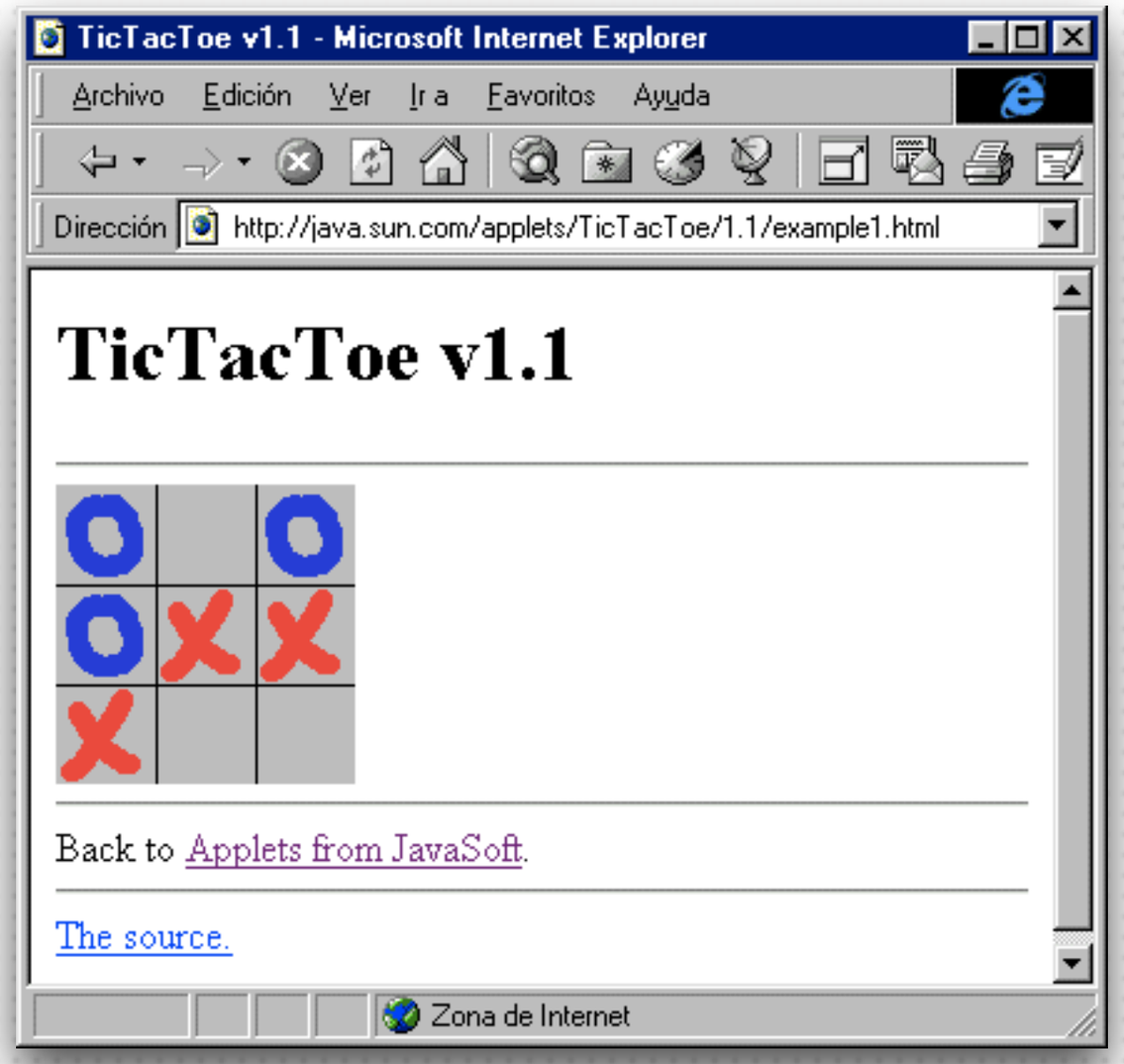
Static Web Content



Web Applets



Java Applet

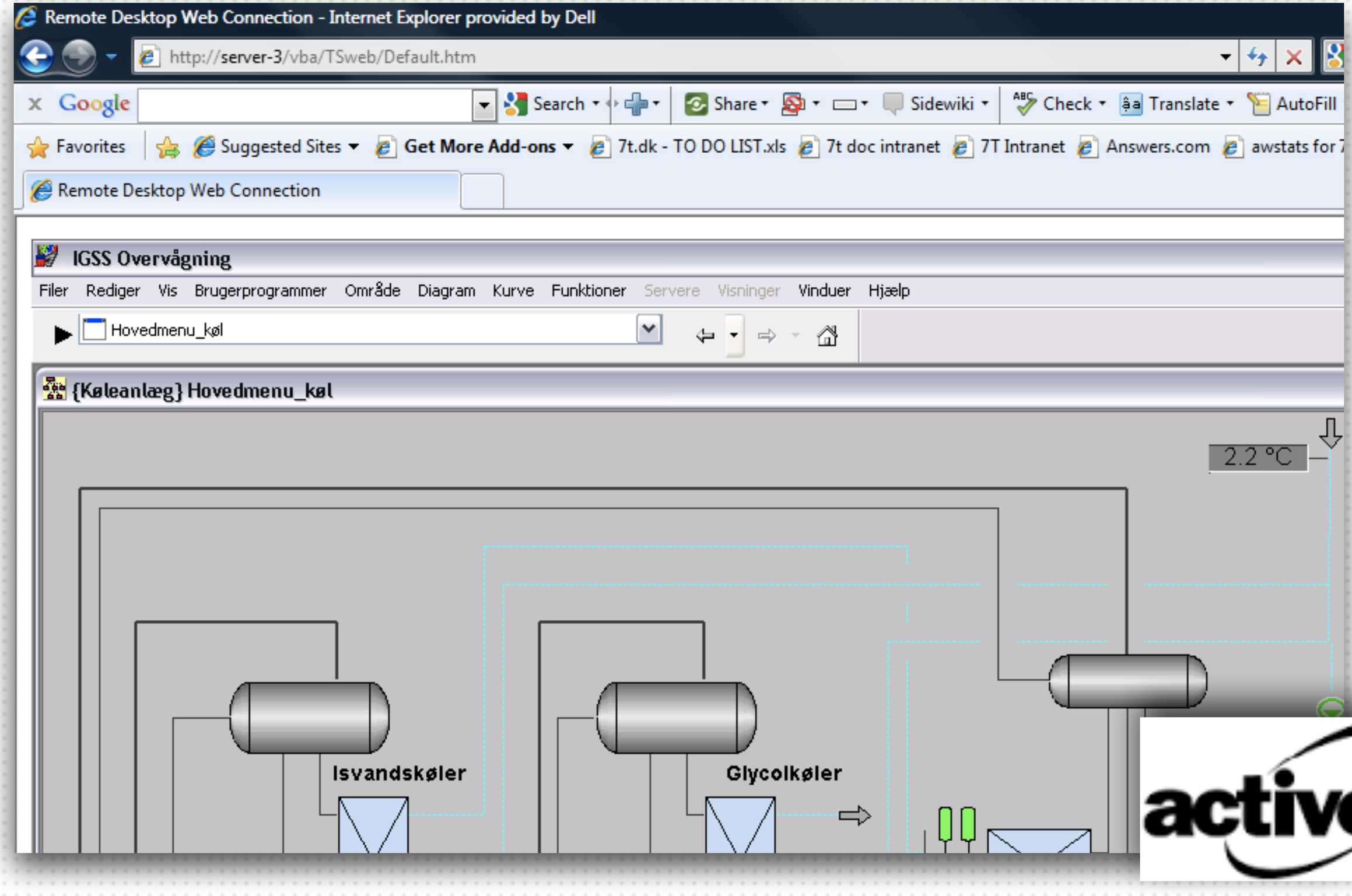




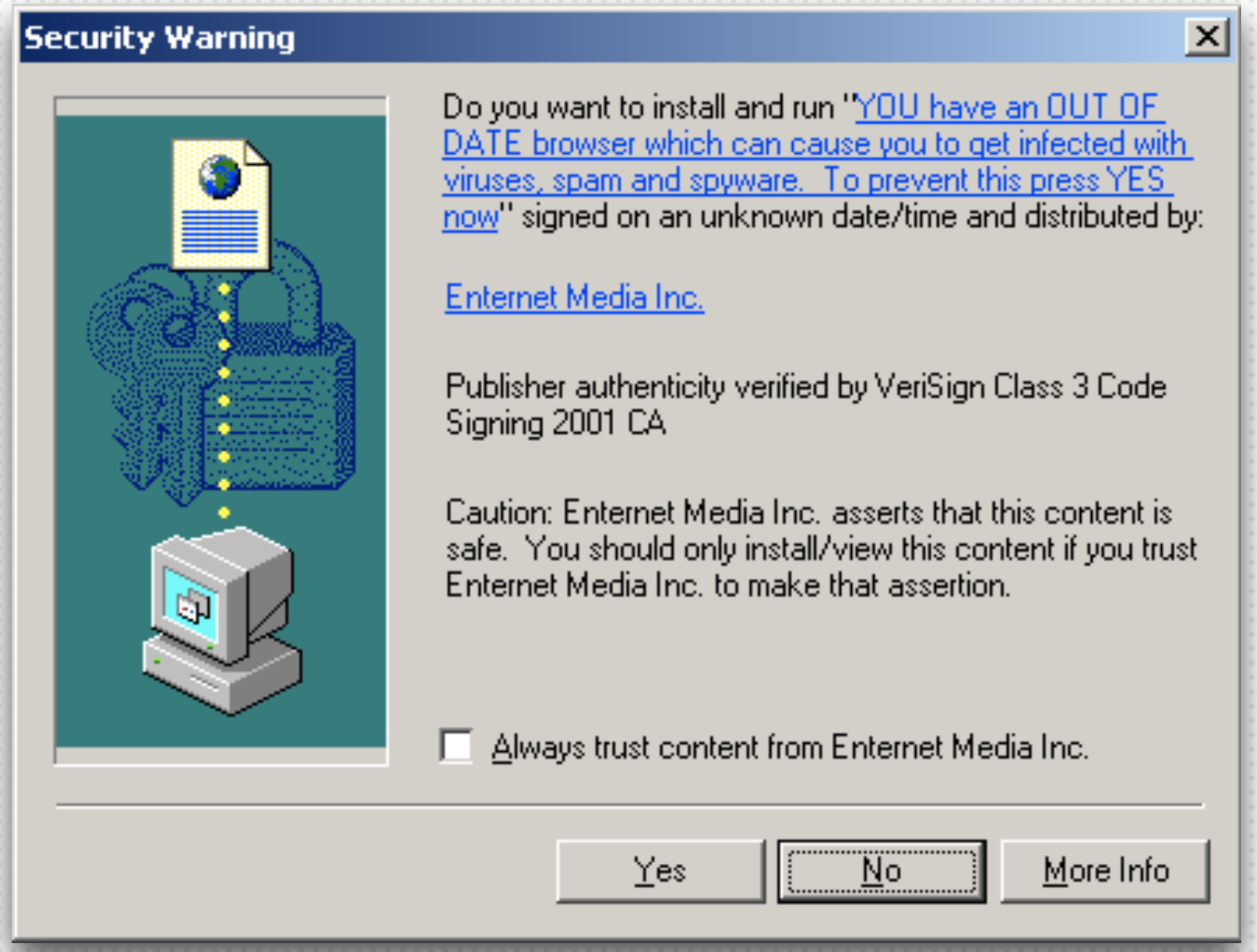


Java Applets

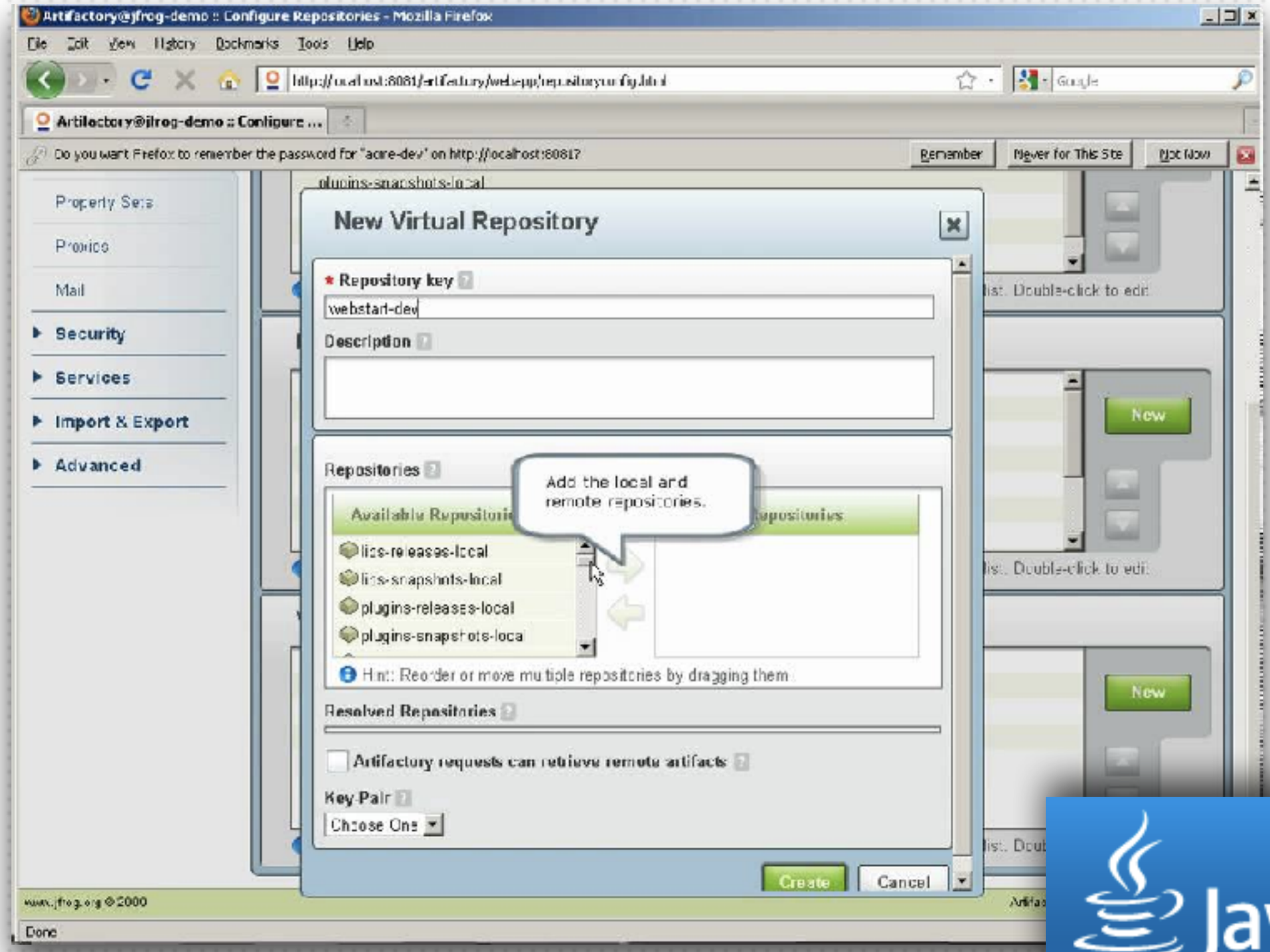
Microsoft ActiveX



ActiveX Security Warning



JavaFX



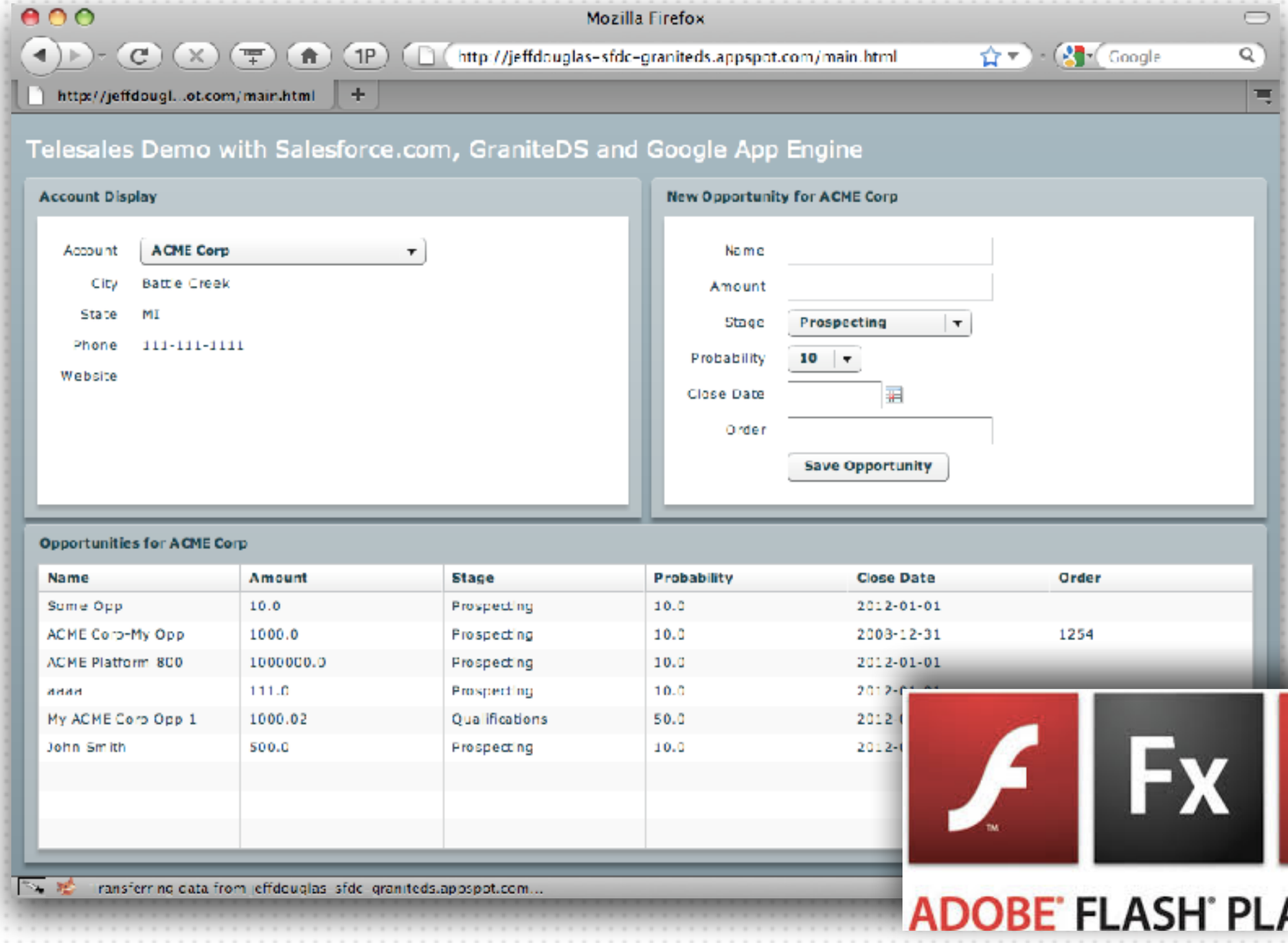
JavaFx

Microsoft Silverlight

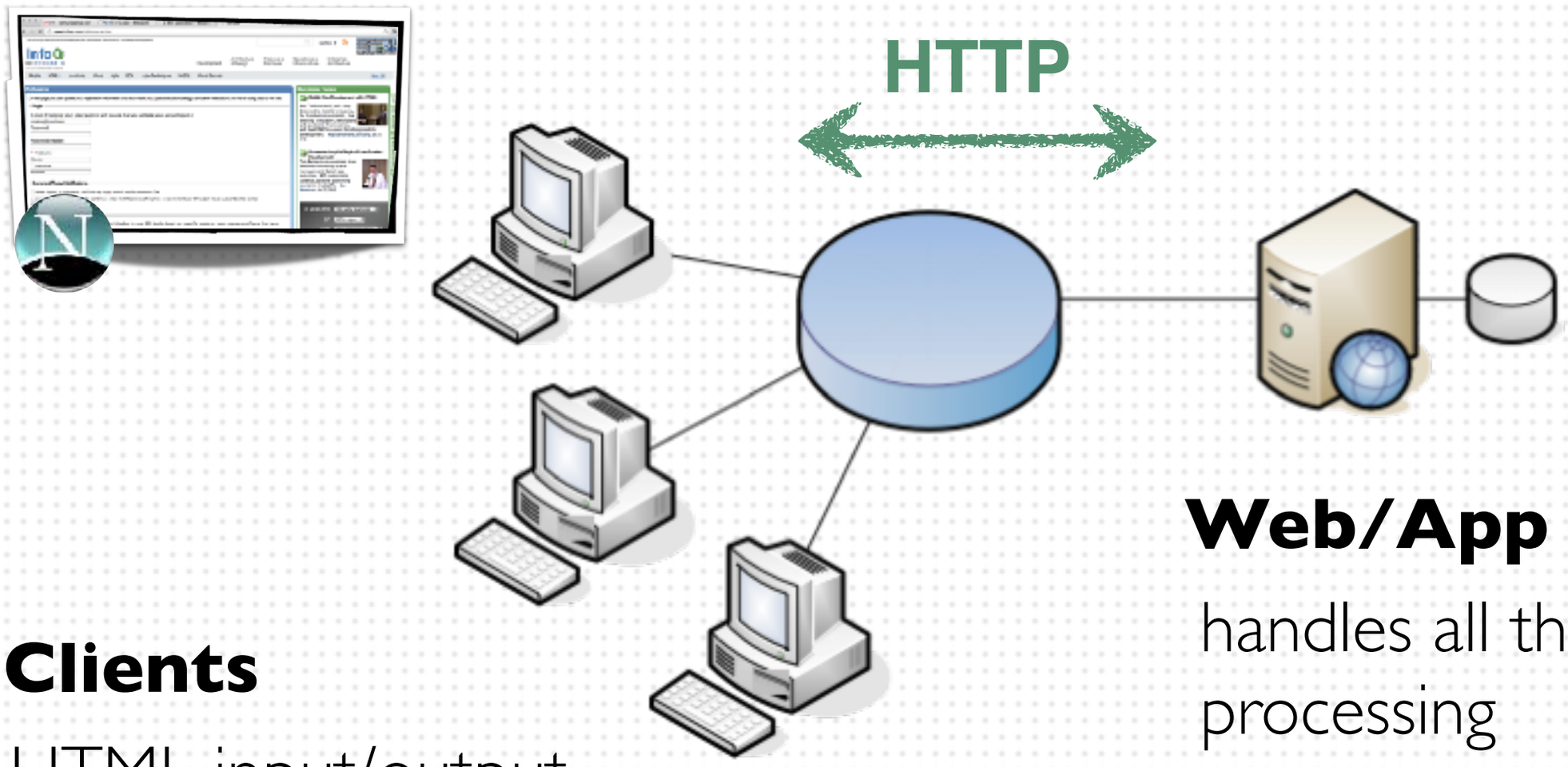


Microsoft® Silverlight™

Adobe Flex


Mozilla Firefox
http://jeffdouglas-sfdc-graniteds.appspot.com/main.html
Google
Telesales Demo with Salesforce.com, GraniteDS and Google App Engine
Account Display
Account
ACME Corp
City
Battle Creek
State
MI
Phone
111-111-1111
Website
New Opportunity for ACME Corp
Name
Amount
Stage
Prospecting
Probability
10
Close Date
Order
Save Opportunity
Opportunities for ACME Corp
Name
Amount
Stage
Probability
Close Date
Order
Some Opp
10.0
Prospecting
10.0
2012-01-01
ACME Corp-My Opp
1000.0
Prospecting
10.0
2008-12-31
1254
ACME Platform 800
1000000.0
Prospecting
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111.0
Prospecting
10.0
My ACME Corp Opp 1
1000.02
Qualifications
50.0
John Smith
500.0
Prospecting
10.0
Fx
ADOBE FLASH PLATFORM

Web Applications



Clients

HTML input/output through web browsers

personal computers running a web browser

Web/App Servers

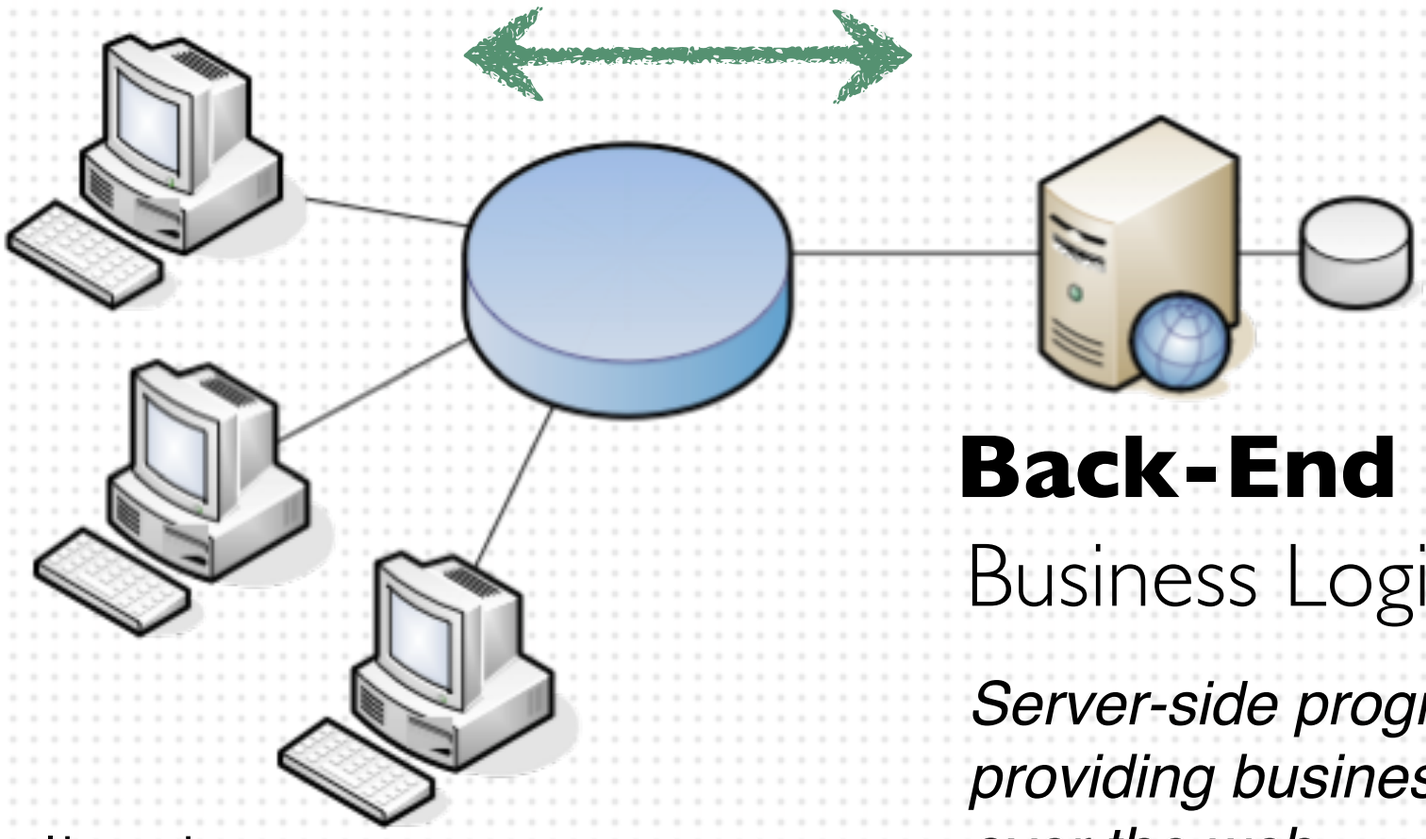
handles all the processing

mainframe and mid-range computers

Rich Internet Applications



Web API
(SOAP/REST/POX/...)



Rich Clients

Single-Page Application
(JavaScript-HTML5-CSS)

Single-page JavaScript program running inside web browser

Back-End Servers

Business Logic Services

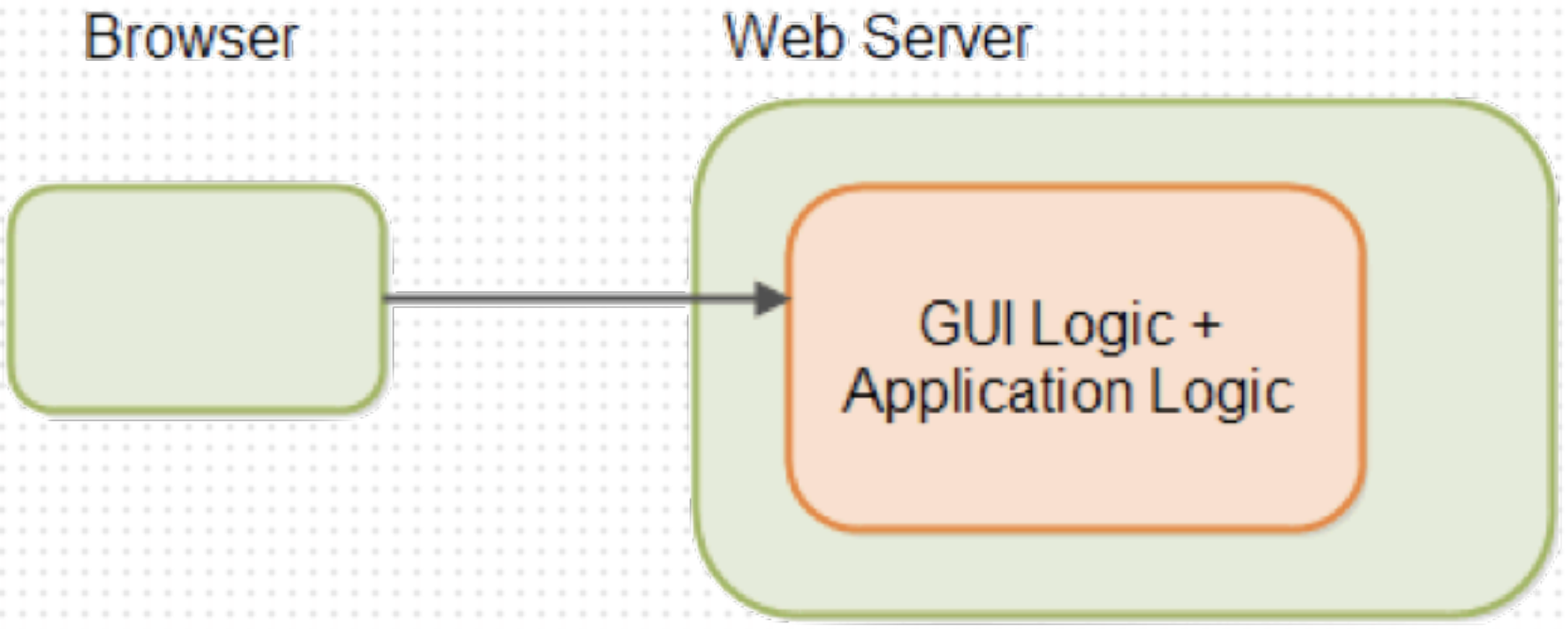
Server-side programs providing business services over the web



RIA Architecture Evolution

http://tutorials.jenkov.com/software-architecture/ria-architecture.html

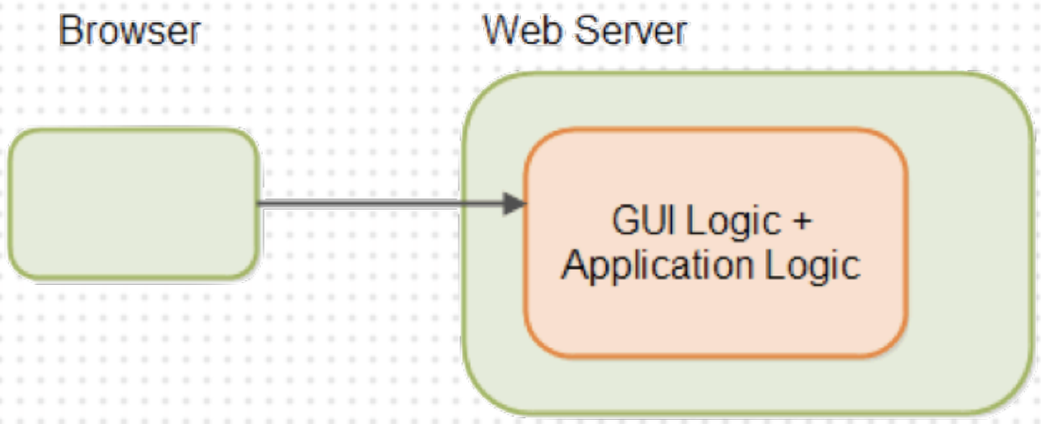
The First Generation Web Apps



Example Technologies:

- JSP
- ASP
- PHP
- CGI
- …

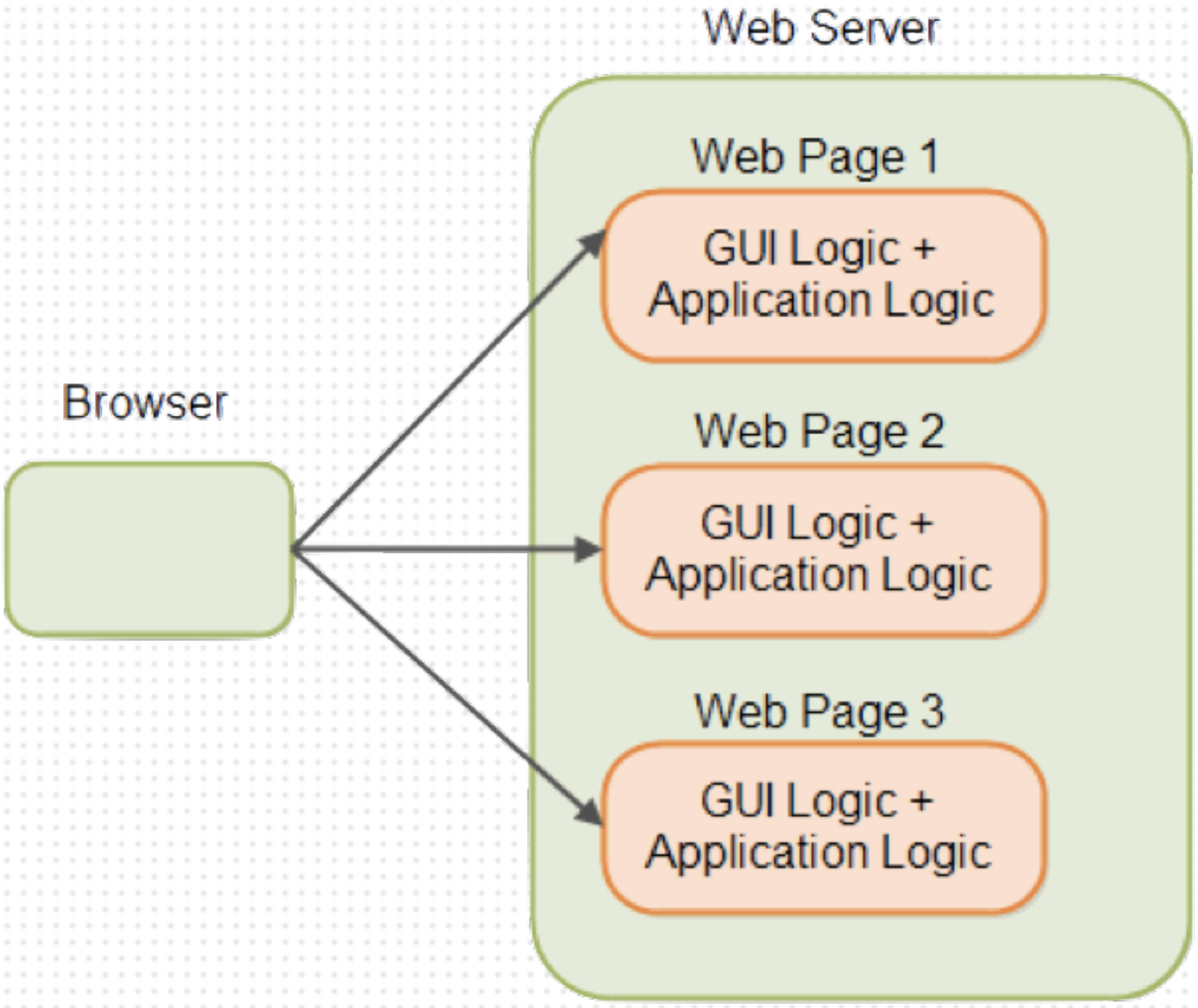
The First Generation Web Apps



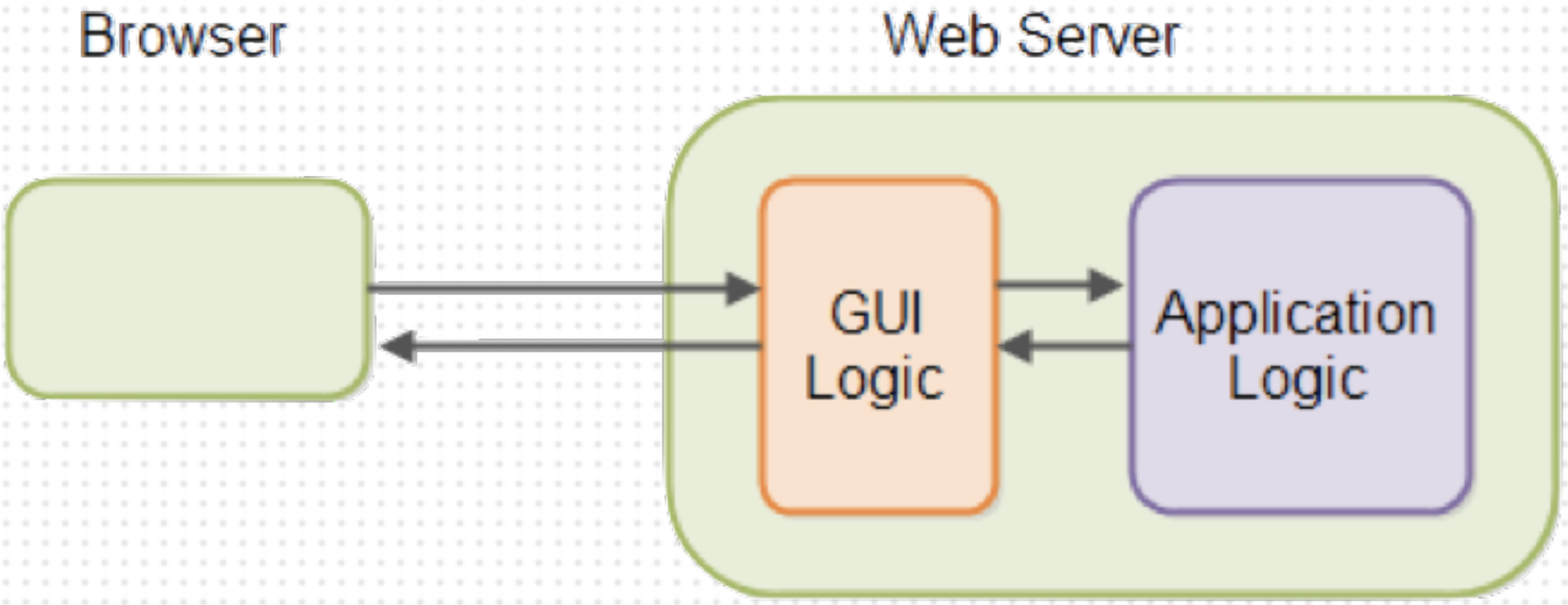

```jsp
<jsp:include page="header.jsp" />

<%
   String firstName = request.getParameter("firstName");
   String lastName = request.getParameter("lastName");
   String destination = request.getParameter("destination");
   String numberOfTickets = request.getParameter("numberOfTickets");
%>
<h1>The flight is booked successfully!</h1>
In flight to
<%=destination%>,
<%=numberOfTickets%>
seats are reserved for
<%=firstName%>
<%=lastName%>.
```

Page-Based Structure



The Second Generation Web Apps



Example Technologies:

- JSF
- ASP.NET
- Spring MVC
- …

The Second Generation Web Apps

```
<div class="<%=classValue%>">
    <label class="control-label" for="inputEmail4">First Name</label> <input type="text"
    class="form-control" name="firstName" id="inputEmail4" placeholder="First Name">
    <%if(request.getParameter("badFirstName") != null) {%>
        <span class="glyphicon glyphicon-remove form-control-feedback"></span>
    <%}%>
</div>
```

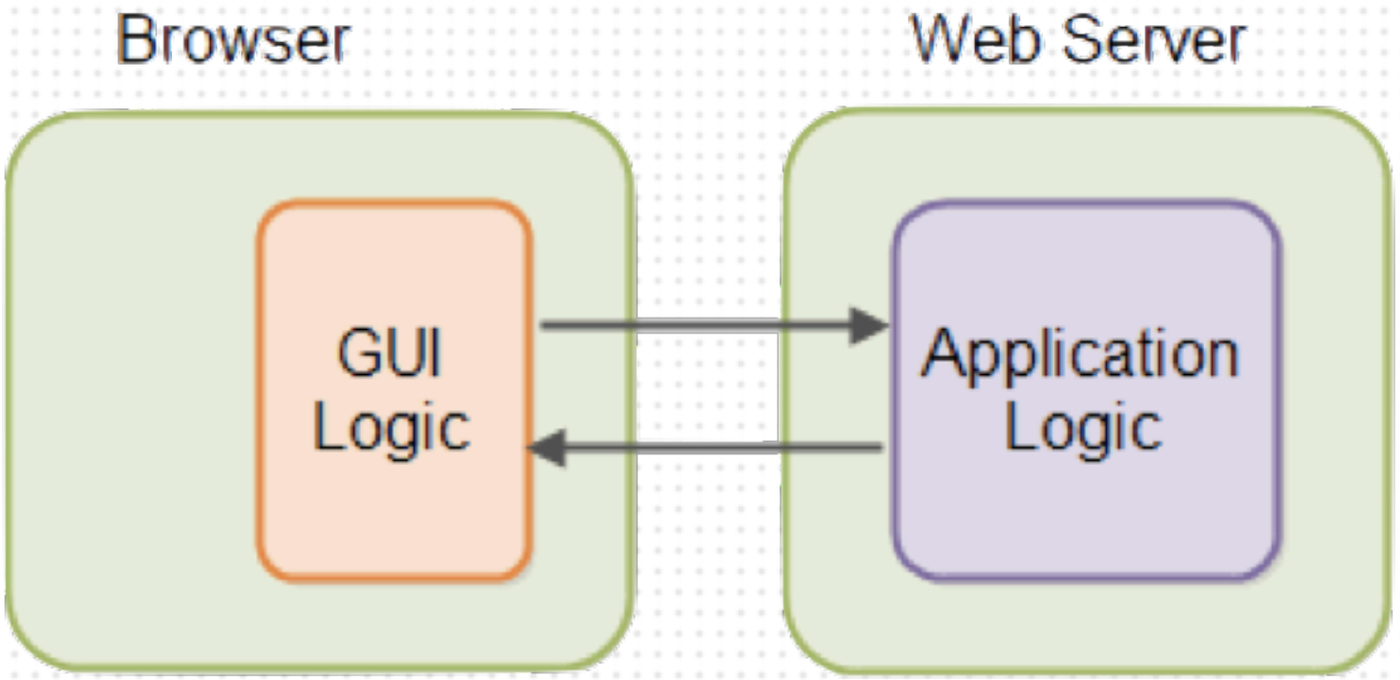
```
@WebServlet("/bookFlight")
public class BookFlightController  extends HttpServlet {
    public void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        String firstName = request.getParameter("firstName");
        String lastName = request.getParameter("lastName");
        String destination = request.getParameter("destination");
        String numberOfTickets = request.getParameter("numberOfTickets");

        if(StringUtils.isBlank(firstName) || StringUtils.isBlank(lastName) ||
              StringUtils.isBlank(destination) || StringUtils.isBlank(numberOfTickets)) {
            String buyPageName = "/buy.jsp";
            if(StringUtils.isBlank(firstName))
              request.setAttribute("badFirstName", "true");
            if(StringUtils.isBlank(lastName))
              request.setAttribute("badLastName", "true");

            RequestDispatcher rd = getServletContext().getRequestDispatcher(buyPageName);
            rd(request, response);
        } else { ... }

    }
}
```

RIA Web Apps



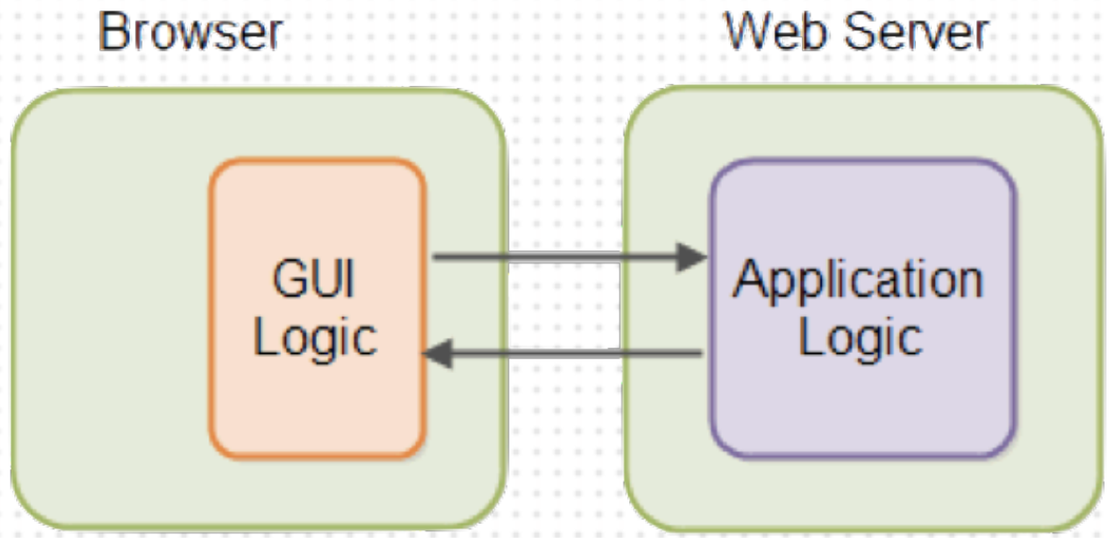
Example Technologies:

- JS+HTML5+CSS3
- GWT

- Flex
- JavaFX
- Silverlight

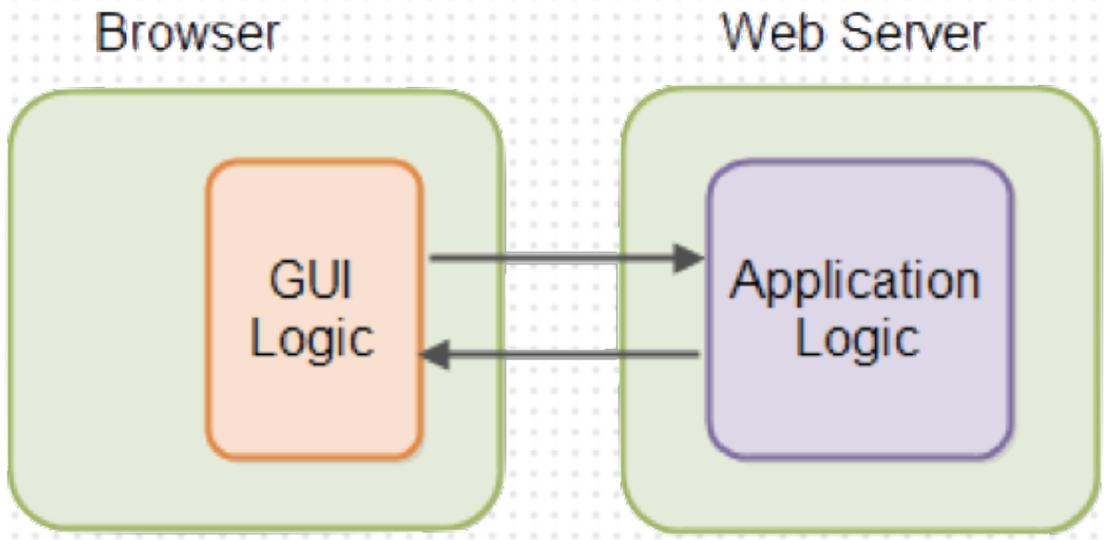


RIA Consequence



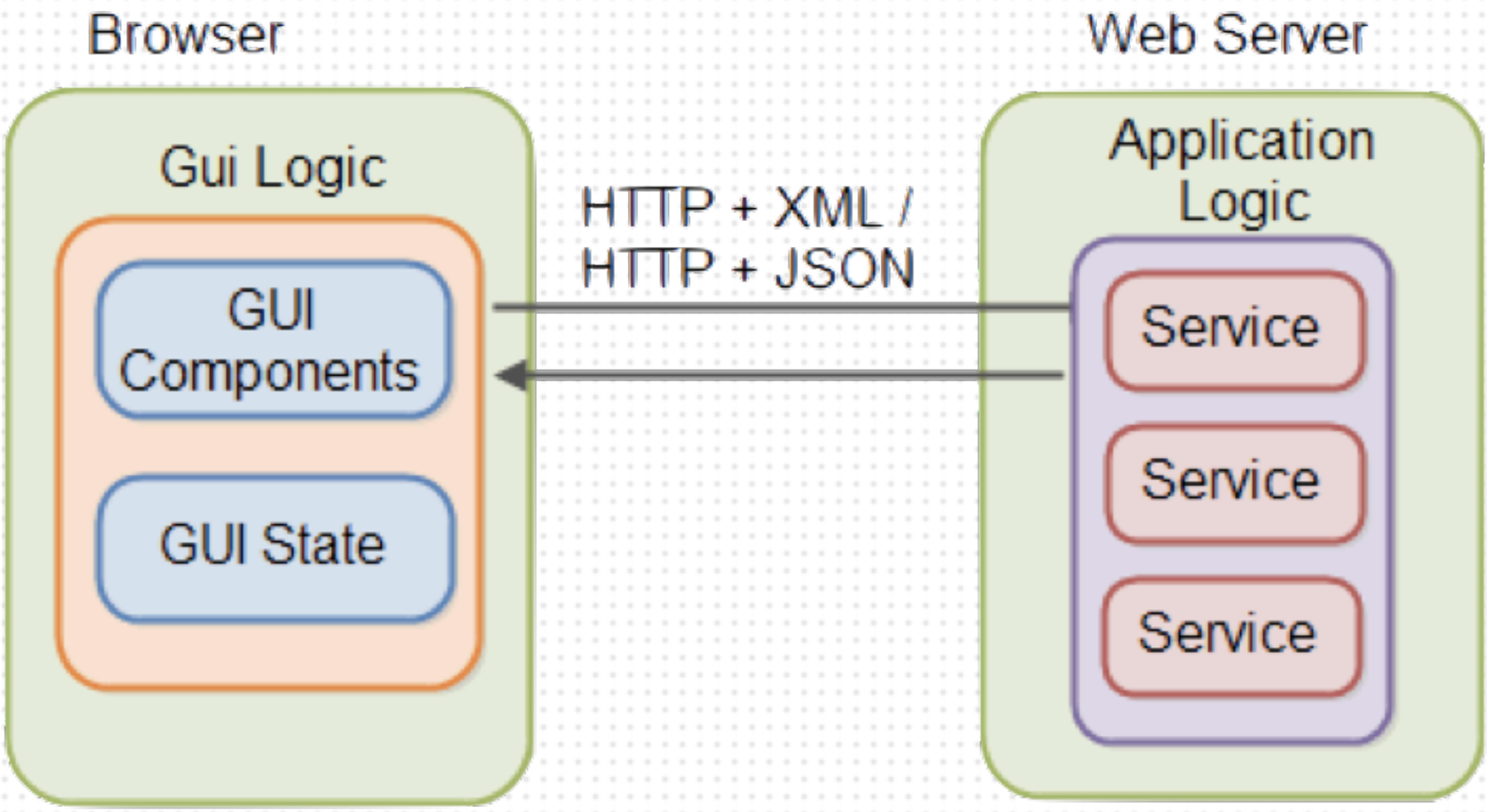
1. Much more advanced GUIs
2. Freeing up CPU from the server side
3. The GUI state is kept in browsers
4. Only exchanging data → Lower network traffic
5. Reusable GUI components

RIA Consequence



6. Server side codes are structured around services, not pages
7. GUI is independent from the server side code language
8. Independent development of GUI and server side codes
9. Adding different types of client is easy
10. Providing API access to server side is easy
11. Facilitating service testing (server side codes testing)

Get a Closer Look at RIA Web Apps



```java
@RestController
@RequestMapping("/flights")
public class FlightService {
    @Autowired
    FlightManager flightManager;

    @GetMapping(value = "/{destination}", produces = MediaType.APPLICATION_JSON_VALUE)
    public FlightInfo getAvailableSeats(
            @PathVariable(value = "destination") String destination) {
        //Validation is required
        return FlightRepository.getInstance().getAvailableSeats(destination);
    }

    @GetMapping(value = "", produces = MediaType.APPLICATION_JSON_VALUE)
    public Collection<FlightInfo> getflights() {
        //Validation is required
        return FlightRepository.getInstance().getFlights();
    }

}
```

```java
@RestController
@RequestMapping("/flights")
public class FlightService {
    @Autowired
    FlightManager flightManager;

    @GetMapping(value = "/{destination}", produces = MediaType.APPLICATION_JSON_VALUE)
    public FlightInfo getAvailableSeats(
            @PathVariable(value = "destination") String destination) {
        //Validation is required
        return FlightRepository.getInstance().getAvailableSeats(destination);
    }

    @GetM
    publi
        //
        re
    }

}
```

```java
@RestController
@RequestMapping("/reservations")
public class ReservationService {

    @Autowired
    FlightManager flightManager;

    @PostMapping
    public TicketInfo bookTheFlight(
            @RequestBody BookingRequestInfo requestInfo) {
        //Validation is required
        TicketInfo ticketInfo = flightManager.bookFlight(
                requestInfo.getDestination(), requestInfo.getNumberOfTickets());
        ticketInfo.setFirstName(requestInfo.getFirstName());
        ticketInfo.setLastName(requestInfo.getLastName());
        return ticketInfo;
    }
}
```

Demo